

AI DDR Report Generation - System Architecture

Stage	Purpose	Output
1. Document Ingestion	Accept inspection and thermal reports in PDF or DOCX format.	Raw document objects
2. Text Extraction	Extract observations, areas, readings, and notes.	Structured text chunks
3. Image Extraction	Extract visual and thermal images from source reports.	Tagged evidence images
4. Evidence Merge	Combine duplicate or related inspection and thermal findings.	Consolidated area-wise observations
5. Validation	Mark missing or conflicting details clearly.	Reliability notes
6. DDR Writer	Generate client-friendly report with severity and recommendations.	Final DOCX or PDF report

Reliability Design

The system avoids inventing facts. If a field is missing, it writes Not Available. If an image is expected but not found, it writes Image Not Available. If inspection and thermal findings conflict, the final report mentions the conflict instead of hiding it.

Deployment Option

The included Streamlit app can be deployed on Streamlit Community Cloud or converted into a FastAPI service for production use.